KD5 differs from KT4 (which equals S4)

The Open Logic Project contributors

openlogic:normal-modal-logic/axioms-systems:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the load-bearing countermodel figure fig:KD5not4 (source lines 1614-1637) sits outside the proof environment in the source; it is delivered verbatim as a context span (LANE_BRIEF 15.5/15.7, ren precedent), and the proof cites it via olref which renders as an internal reference to the delivered figure

Editorial note. The label thm:KD5not4 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Soundness of schemas. If a relation R has one of the listed properties (serial, reflexive, symmetric, transitive, euclidean), every instance of the corresponding axiom schema (D, T, B, 4, 5) is true in every model with that relation (thm:soundschemas, sibling frame-definability unit).. Opens the proof: all instances of D and 5 are true in serial euclidean models, so by soundness a countermodel need only be serial euclidean and invalidate one instance of 4.

Source reference. Original theorem thm-soundschemas: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1606–1606. Named citation invoked at source line(s) 1606: the author's marker \olref[frd][acc]{thm:soundschemas} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. KD5 $\neq$ KT4 = S4.

Proof. By thm-soundschemas we know that all instances of D and 5 are true in all serial euclidean models. So it suffices to find a serial euclidean model containing a world at which some instance of 4 fails. Consider the model of 1 , and notice that $/ \models M[\Box p \to \Box\Box p][w_1]$. □

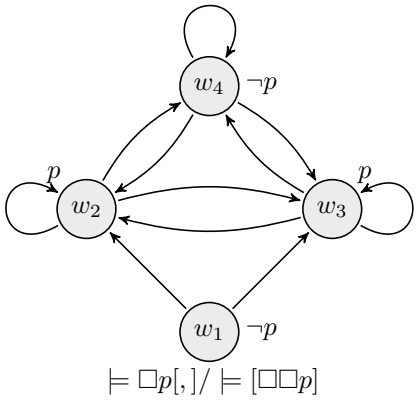

Figure 1: The model for 0.1 .